

idea

PROVENANT

An Autonomous Financial Agent That Tests Two Questions: "Does this decision still deserve to exist?" and "Does this strategy still deserve to exist?" — Enforced in Software, and Physically Sealed in Hardware.

CSI ORIGIN 2026 — Problem Statement 3: Autonomous AI Agents for Real-Time Financial Markets

1. ONE-PARAGRAPH CONCEPT

Provenant is an autonomous financial agent built around two connected feedback loops instead of one prediction model. The **Inner Loop** (Decision Validity Engine) makes trading decisions from explicit, typed evidence — price, liquidity/order-book, volatility, news — and continuously recomputes a live Validity score as that evidence decays, gets contradicted, or loses source trust; when Validity breaks an adaptive threshold, the agent autonomously holds, reduces, pauses, cancels, or reverses the position and explains exactly which evidence broke. The **Outer Loop** (Research Sleeve) watches for *recurring* invalidation patterns ("news+momentum decisions keep failing during high-volatility earnings windows"), and in response autonomously generates a new hypothesis from a structured, bounded template space, backtests it, walk-forward validates it, paper-trades it, and only promotes it into the live strategy pool if it clears a statistical gate — all inside a capital firewall so an experimental idea can never touch real allocation until proven. The connective tissue between the two loops — and the part a normal AI agent has none of — is physically enforced by an **M5StickC Plus2** acting as a temporal commit–reveal lock: before the Research Sleeve is allowed to see a candidate strategy's backtest result, the exact hypothesis, parameters, and test window must be SHA-256 committed and locked on the device for a fixed window, so the agent cannot see a bad result and quietly retry with tweaked parameters (p-hacking). This makes both the *individual decision* and the *research process that produces new decisions* auditable, adaptive, and honest — not a bigger prediction model, but a different architecture for how belief is formed, revised, and improved.

2. PRODUCT NAME

PROVENANT — provenance (every decision traces to its evidence) + covenant (the hard boundaries humans set and the commitment the agent can't wriggle out of). Sounds like a

real fintech infra product, not a generic "AI-something" name. Subsystems keep the brand consistent:

Component	Name
Inner loop (decision-level)	Decision Validity Engine (DVE)
Outer loop (strategy-level)	Research Sleeve
Hardware commit–reveal device	The Vault (running on M5StickC Plus2)

3. THE TWO-LEVEL LEARNING SYSTEM, IN PLAIN TERMS

Decision-Level Learning (Inner Loop) — "Do I still believe this trade?"

Every decision is not a verdict, it's a claim backed by evidence, and the claim has an expiry condition. If the evidence that justified a BUY decays, gets contradicted, or its source turns out to be unreliable, the system stops believing the decision *before* a stop-loss would ever trigger — and says exactly why ("40% of this decision depended on liquidity conditions that no longer exist, 35% on news that's now contradicted").

Strategy-Level Learning (Outer Loop) — "Does my current playbook still work?"

A single bad trade is not a research signal. But if the *same kind* of decision keeps getting invalidated under the *same* conditions (e.g., news+momentum entries during high-vol earnings windows), that's a pattern worth investigating. The Research Sleeve treats this as a hypothesis-generation trigger: it proposes a bounded, structured alternative (not a blank-slate "invent any strategy"), tests it rigorously offline, and only if it survives out-of-sample validation does it get promoted into the live strategy pool — completely isolated from real capital until then.

Why connecting them is more powerful than either alone:

A system with only the Inner Loop gets better at *exiting* bad trades but never gets better at *avoiding* them — it will make the same mistake forever, just cut losses faster each time. A system with only the Outer Loop (periodic strategy re-optimization) has no fast, real-time defense against an individual decision going bad *right now* — it's too slow to react intraday. Chaining them means individual failures aren't just absorbed and forgotten, they *accumulate into evidence* that drives a structured self-improvement process — and only rigorously-tested improvements make it back into live decisions, which the Inner Loop will then, again, hold to the same standard of continuous justification.

4. THE ONE-LINE DIFFERENTIATOR

- **Technical:** "A decision-validity graph with per-evidence decay drives real-time invalidation, and recurring invalidation patterns trigger a capital-isolated, commit-reveal-locked hypothesis testing pipeline before any strategy is promoted."
- **Judge-friendly:** "The agent doesn't just trade — it continuously checks whether it still believes its own trades, and when it notices it keeps being wrong for the same reason, it runs a locked, tamper-proof experiment to find something better, before ever risking real capital on the idea."
- **Extremely simple:** "It asks 'should I still believe this?' about every trade, and 'should I still trust this strategy?' about its whole playbook — and it can't cheat on either answer."

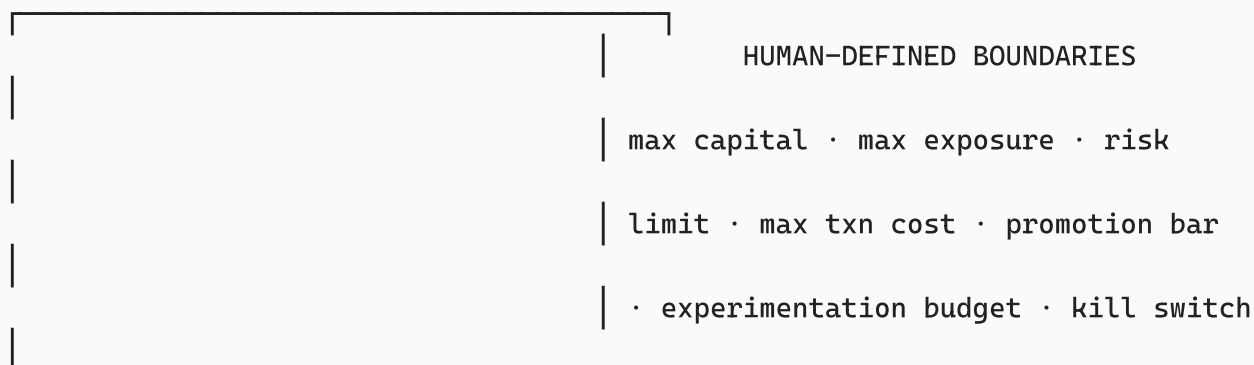
5. HOW THIS MAPS TO THE OFFICIAL PROBLEM STATEMENT

The CSI ORIGIN 2026 brief (Problem Statement 3) asks for a continuous OBSERVE → INTERPRET → REASON → EVALUATE RISK → ALLOCATE → EXECUTE → OBSERVE OUTCOME → ADAPT loop, explicitly rejecting price-prediction-only systems and fixed-rule bots. Provenant satisfies this at both levels:

- The **Inner Loop** is the "**continuously reassess decisions**" requirement, made architectural rather than a polling timer.
- The **Outer Loop** is the "**adapt / incorporate outcomes into subsequent decision-making**" requirement, made rigorous rather than "retrain occasionally."
- The **Vault** is the **enforcement layer for "dynamic decision-making rather than a fixed strategy"** — it's what stops the outer loop itself from degenerating into the exact fixed/overfit-strategy trap the brief is warning against.

Full requirement-by-requirement mapping is in Section 7.

6. COMPLETE SYSTEM ARCHITECTURE



stage below

enforced at every

REAL-TIME DATA INGESTION

price ticks · order-book/liquidity proxy · volatility ·
news/events



MARKET INTERPRETATION

regime detection · feature synthesis · freshness tagging



EVIDENCE / ASSUMPTION LAYER

EvidenceNode{type, source, timestamp, decay_fn, trust_score}

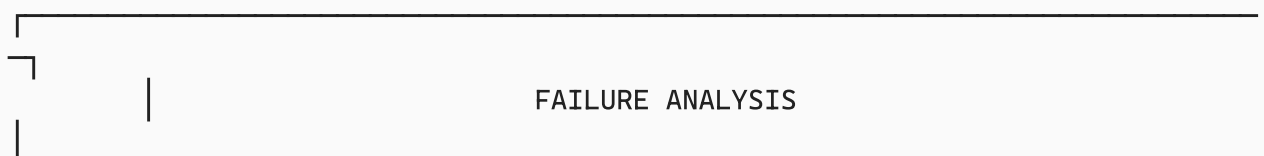
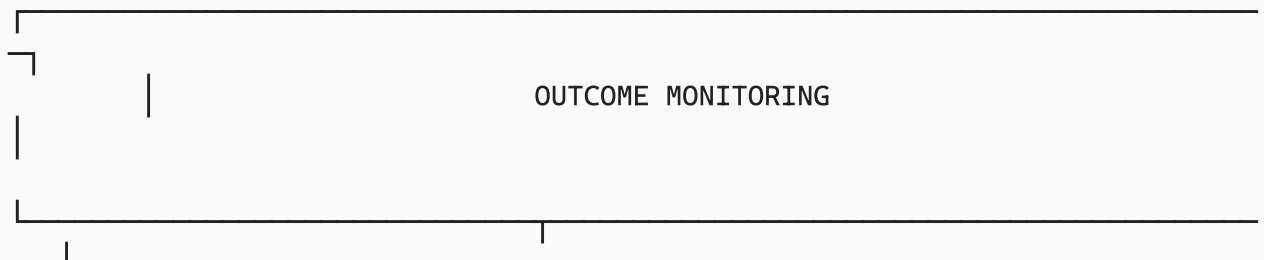
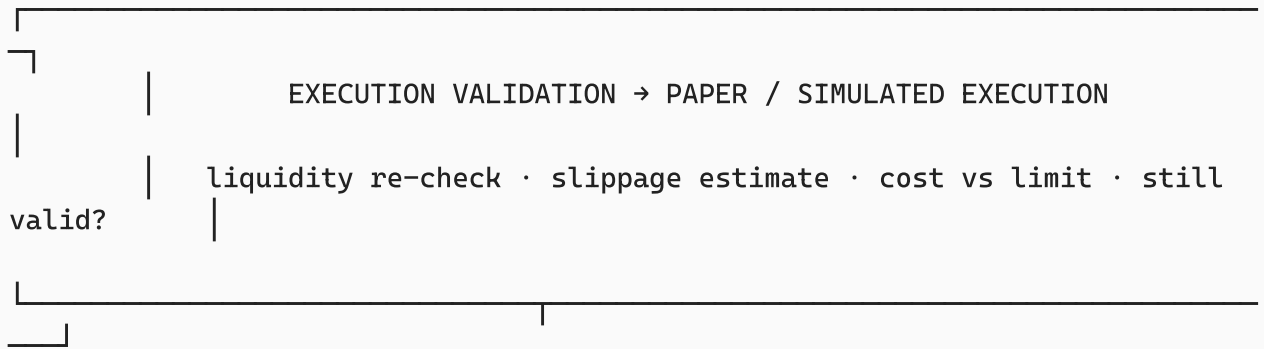
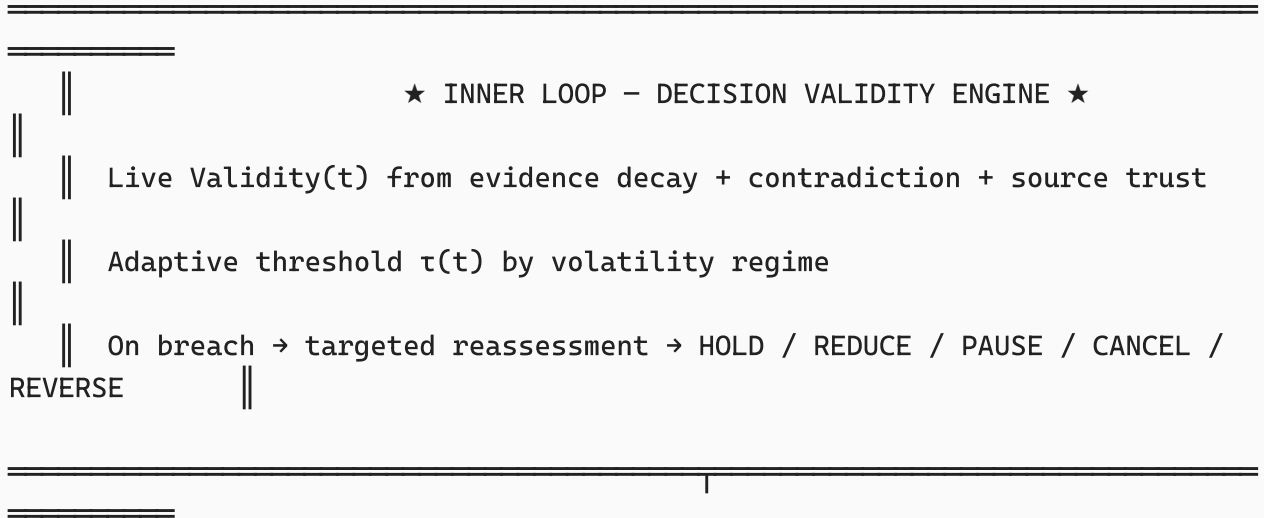
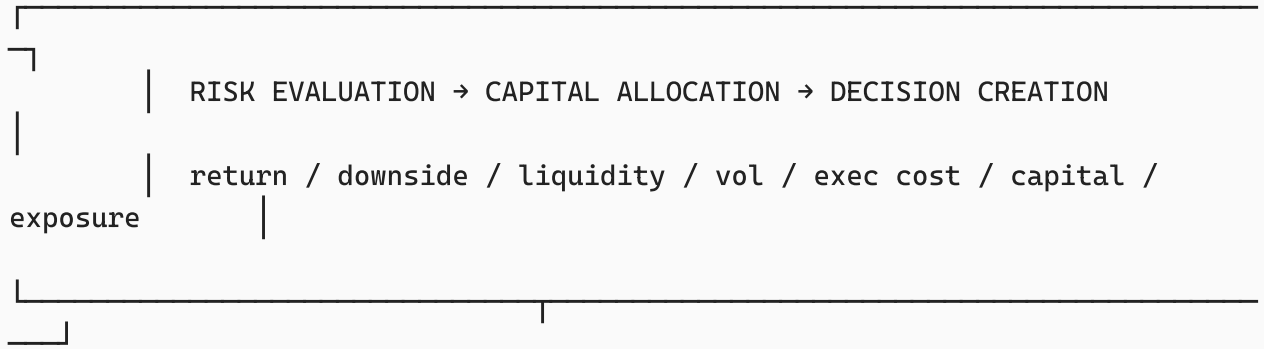
FRESH / DEGRADING / STALE classification per evidence type



OPPORTUNITY GENERATION

draws from current STRATEGY POOL (bottom of diagram) + live
evidence





tags WHY a decision was invalidated (evidence type, source)



RECURRING PATTERN DETECTION

groups failures by {evidence type × regime × strategy template}

crosses threshold (e.g. 3+ similar failures) → RESEARCH TRIGGER



★ OUTER LOOP – RESEARCH SLEEVE (CAPITAL-FIREWALLED) ★

HYPOTHESIS GENERATION (structured template space, not open-ended)



🔒 THE VAULT – M5StickC Plus2 COMMIT-REVEAL

canonical experiment → SHA-256 commit → LOCKED 60s

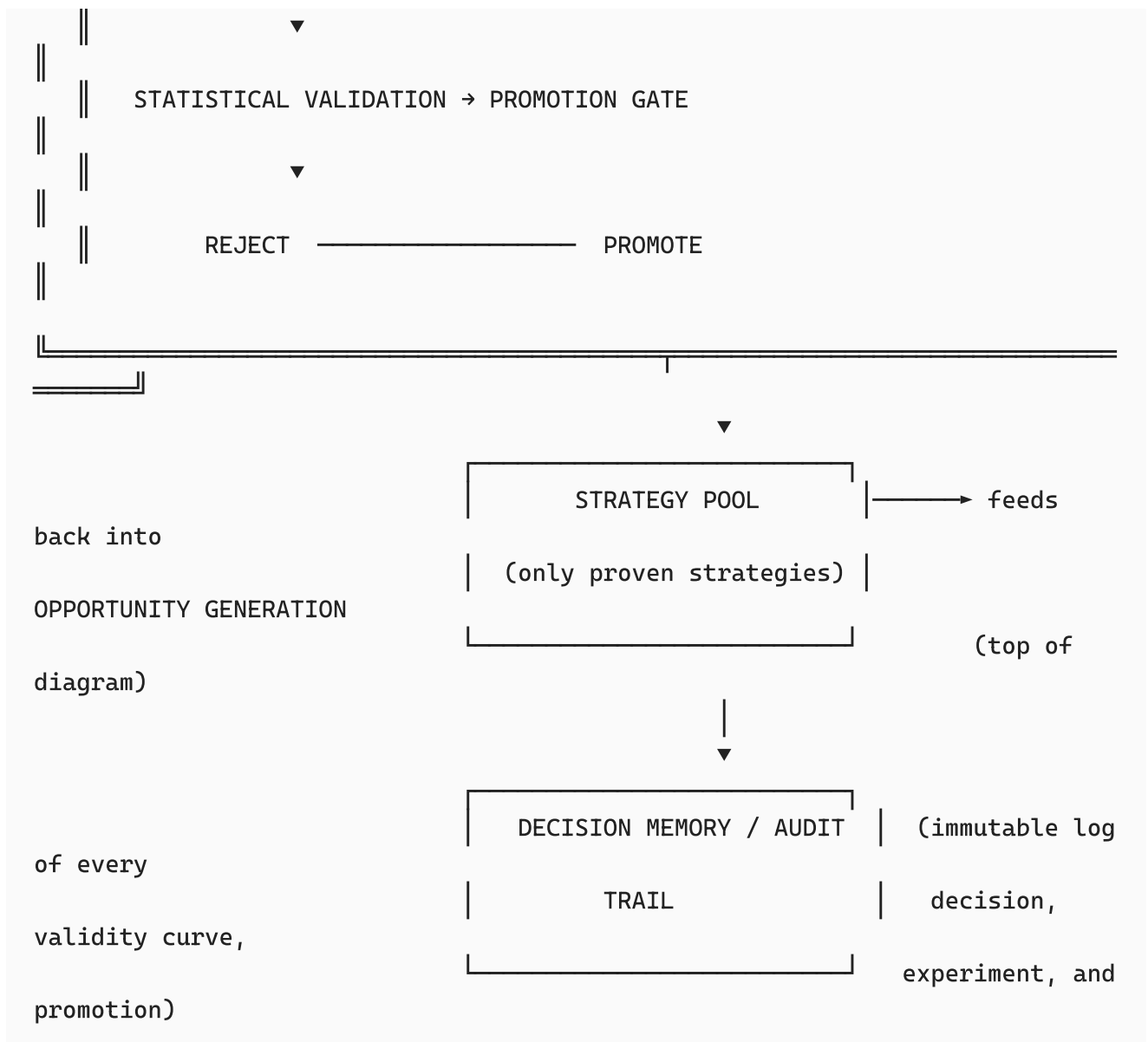
backend independently enforces lock (device is a
physical witness, not the source of truth)

▼ (only after lock expires)

BACKTEST → OUT-OF-SAMPLE VALIDATION → PAPER TEST



REVEAL: result_hash = SHA256(commit + result); verify vs stored
commit



Firewall: the Research Sleeve reads market data and decision-failure history, but has **no write access to live Allocation or Execution**. A promoted strategy only becomes usable by writing one row to `StrategyPool` — which the live Opportunity Generation stage reads on its next cycle. There is no code path from "experiment result" to "live order" that skips the Promotion Gate.

Human boundaries are enforced in exactly three places: (1) Capital Allocation (max capital/exposure), (2) Execution Validation (max txn cost / slippage), (3) Promotion Gate (statistical bar, max experimentation budget). Everything else is autonomous.

7. STEP-BY-STEP END-TO-END FLOW

1. **Market data arrives** — a price tick, an order-book snapshot, and a news headline stream in.
2. **Opportunity created** — using the current Strategy Pool's "momentum + news confirmation" template, the system proposes: *BUY Asset X — positive earnings news*,

strong momentum, high liquidity, buyer imbalance.

3. **Risk evaluated** — volatility-adjusted downside, liquidity-adjusted slippage estimate, exposure impact.
4. **Capital allocated** — sized within max-capital/max-exposure limits, ranked against other open candidates.
5. **Decision created, evidence recorded** — a Decision Node cites the 4 EvidenceNodes above with explicit weights (e.g., news 35%, momentum 25%, liquidity 20%, imbalance 20%) and invalidation conditions.
6. **Conditions change** — 90 seconds later, a corrected headline contradicts the earnings news, and order-book imbalance normalizes.
7. **Decision becomes invalid** — Validity recomputes from 0.91 to 0.34 (below adaptive threshold τ); Failure Analysis tags this as "news-contradiction-driven, high-vol regime."
8. **Autonomous corrective action** — position is REVERSED; the dashboard shows the exact evidence breakdown.
9. **Similar failures recur** — over the session, 3+ decisions using the "news + momentum" template fail the same way during high-volatility windows.
10. **Research Sleeve activates** — Recurring Pattern Detection fires a Research Trigger: *"news+momentum entries underperform in high-vol regimes — test a confirmation-delay variant."*
11. **Hypothesis generated** — from the structured template space: *"Wait 5 minutes after high-impact news + high volatility + liquidity recovery before entering, vs. immediate entry."*
12. **Commit → Vault → Test** — the canonical experiment object is hashed, sent to the M5StickC Plus2, which displays COMMIT LOCKED and counts down 60s; the backend independently refuses any parameter change during the lock; only after the timer expires does the backtest + walk-forward + paper-test run.
13. **Reveal & validate** — result hash is computed and checked against the stored commit; if it matches and p-value/Sharpe/decay clear the statistical bar, the device shows VERIFIED / PROMOTABLE.
14. **Promotion or rejection** — a losing/overfit hypothesis is rejected and logged (still useful — it rules out an idea); a validated one is promoted into the Strategy Pool.
15. **Future decisions improve** — the next high-vol news event now triggers Opportunity Generation to consider *both* the old template and the new delayed-entry template, and the Inner Loop holds the new template's decisions to the exact same continuous-validity standard.

8. CONSTRAINT COVERAGE (mapped to the official CSI ORIGIN 2026 brief)

Problem Statement Requirement	System Component	Exactly How It's Satisfied
Continuously process heterogeneous real-time information (prices, liquidity, news, signals)	Real-Time Data Ingestion + Evidence Layer	Parallel ingestion of price, order-book/liquidity proxy, volatility, and news streams, each converted to typed EvidenceNodes
Distinguish current vs. no-longer-representative information	Evidence/Assumption Layer	Per-evidence-type decay function classifies FRESH / DEGRADING / STALE continuously, not on a global timer
Evaluate opportunities on return, liquidity, execution cost, volatility, capital, risk	Risk Evaluation + Capital Allocation	Explicit multi-factor scoring before any opportunity is eligible for allocation
Dynamic decisions, not a fixed strategy	Opportunity Generation + Strategy Pool	Opportunities are drawn from a live, evolving Strategy Pool, not a static rule set; the pool itself only changes via validated promotion
Explicit constraints on capital, exposure, risk, transaction cost	Human-Defined Boundaries block	Enforced at Capital Allocation, Execution Validation, and Promotion Gate — the only 3 human touchpoints
End-to-end loop: observe → interpret → reason → risk → allocate → execute → observe outcome → adapt	Full architecture (Section 6)	Traced concretely in Section 7, steps 1–15, with no human step in between
Continuously reassess decisions as conditions/information change	Decision Validity Engine (Inner Loop)	Live Validity(t) recomputation triggers automatic reassessment on decay, contradiction, or trust change
Opportunity ≠ executable action; consider liquidity, slippage, cost, impact	Execution Validation stage	Independent pre-fill re-check of liquidity, slippage, and cost limits, can defer/block even an approved decision
Incorporate outcomes of previous actions into subsequent decision-making	Outcome Monitoring → Failure Analysis → Research Sleeve	Every outcome (success, invalidation, execution failure) feeds both immediate learning (trust/decay updates) and long-run learning (strategy promotion)

Problem Statement Requirement	System Component	Exactly How It's Satisfied
Minimise human intervention; only human-defined boundaries govern behavior	Entire loop	Humans configure limits once; every HOLD/REDUCE/CANCEL/REVERSE, every hypothesis, every promotion decision is autonomous
System must not be a fixed/overfit strategy dressed up as adaptive (implicit anti-p-hacking requirement of "dynamic, not fixed")	The Vault (commit–reveal hardware)	Physically and cryptographically prevents the one failure mode that turns "autonomous experimentation" into a fixed strategy discovered by cheating — parameter-tweaking after seeing results

9. THE WOW DEMO (12 phases, ~2.5 minutes)

1. Dashboard shows live prices for 2–3 assets; a BUY decision is created with 4 visible evidence nodes, Validity = 0.91.
2. Injected event: a corrected/contradicting headline fires.
3. Validity gauge visibly collapses (0.91 → 0.34) live on screen.
4. System autonomously REVERSES the position; audit panel explains why, citing the exact evidence node.
5. This is fast-forwarded (pre-seeded history) to show 2 more similar failures under the same regime.
6. Recurring Pattern Detection panel lights up: *"3 failures: news+momentum, high-vol regime."*
7. Research Sleeve activates — a hypothesis card appears: *"Test: 5-min confirmation delay vs. immediate entry."*
8. **The M5StickC Plus2 physically shows COMMIT LOCKED**, hash, and a visible 60-second countdown — the room watches the device, not just the screen.
9. Timer hits zero → device flips to REVEALING → backend runs the backtest/OOS/paper-test.
10. Device shows the verification: COMMIT ✓ / RESULT ✓ / VERIFIED — with OOS Sharpe and p-value.
11. Dashboard shows PROMOTED — the new strategy enters the pool.
12. A new high-vol news event is (re-)injected; the dashboard shows the *new* delayed-entry template being used instead of the old one — closing the loop live.

10. WHAT MAKES THIS DIFFERENT

	Traditional Trading Bot	Typical "AI Agent"	Provenant
Who creates strategies?	Human, fixed at design time	Human-authored agent prompts	Agent, from a bounded template space, only after passing the Vault + statistical gate
Can decisions be invalidated?	Only via stop-loss/take-profit	Rarely, usually a timer-based re-run	Continuously, via live Validity(t)
Remembers why it acted?	No	Sometimes logs output, not reasoning	Full evidence-weighted causal trail per decision
Detects recurring reasoning failures?	No	No	Yes — Recurring Pattern Detection is a first-class stage
Can it autonomously experiment?	No	Rarely, and usually unconstrained	Yes, from a structured hypothesis space only
How are experiments isolated?	N/A	Usually not isolated	Architectural firewall — no code path from experiment to live capital except the Promotion Gate
Anti-p-hacking guarantee?	N/A	None	Physically enforced commit–reveal lock (M5StickC Plus2 + backend-independent enforcement)
How does learning happen?	Doesn't	Retrain on schedule	Two connected loops: fast (evidence trust/decay updates) + slow (strategy promotion)

11. NOVELTY SELF-CHECK

"Could a judge say this is just an AI trading bot with extra features?"

No — the two things that would make it "just a bot with a validity score" or "just a bot that backtests" are (a) treating invalidation as a bolt-on monitor rather than the gate every decision must pass through, and (b) treating research as free-form and un-firewalled. Both are explicitly prevented architecturally here: no decision exists outside the Validity Engine, and no experiment can influence capital without clearing the Vault's tamper-evident commit–reveal and the Promotion Gate.

Strongest anticipated criticism: *"Isn't the hardware just a gimmick bolted onto a software idea?"*

Answer: No — remove the Vault and the Research Sleeve has a real, specific weakness: nothing stops the agent (or a human debugging it live) from seeing a bad backtest and re-running with tweaked parameters, which is exactly the p-hacking failure mode that makes "autonomous strategy discovery" untrustworthy. The M5StickC Plus2 is the one component that makes that failure mode *physically impossible to hide*, not just logically discouraged — which is precisely why it's demoed as a standalone, visible moment (Phase 8–10) rather than a background detail.